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The resistance of a constantan wire

Introduction

Electrical resistance describes how strongly a component opposes the current through it. For a uniform wire, R = rhoL/A, where rho is the resistivity of the material, L the length and A the cross-sectional area. If the temperature of the wire stays constant, the resistance should be directly proportional to the length, so a graph of R against L should be a straight line through the origin with gradient rho/A. Constantan is used because its resistivity changes very little with temperature.

Research question: How does the resistance of a 0.38 mm diameter constantan wire depend on its length between 0.10 m and 0.90 m at room temperature?

Hypothesis

I predict that resistance is directly proportional to length, because each additional section of wire adds the same opposition to the current.

Variables

The independent variable is the length of wire between the crocodile clips, measured with a metre rule. The dependent variable is the resistance, calculated from the potential difference across the wire divided by the current through it. The current is kept below 0.30 A by using a fixed supply voltage of 1.5 V and a protective resistor, so that the wire does not heat up noticeably. The same piece of wire is used throughout so that the diameter and material do not change.

Method

The wire was taped along a metre rule. A digital ammeter (resolution 0.001 A) was connected in series and a digital voltmeter (resolution 0.01 V) in parallel with the section of wire between the clips. For each length the switch was closed for less than five seconds, the readings were recorded and the switch was opened again. Nine lengths from 0.10 m to 0.90 m were used, and each was measured three times, moving the clips off

and back on between repeats.

Results

Length / m ±0.002	V / V ±0.01	I / A ±0.001	R / Ohm
0.100	0.42	0.268	1.57
0.200	0.71	0.231	3.07
0.300	0.92	0.204	4.51
0.400	1.07	0.182	5.88
0.500	1.19	0.163	7.30
0.600	1.28	0.149	8.59
0.700	1.35	0.137	9.85
0.800	1.40	0.126	11.11
0.900	1.44	0.117	12.31

Data analysis

Resistance was calculated with R = V/I for each trial and then averaged. The percentage uncertainty in R was found by adding the percentage uncertainties in V and I. A graph of R against L gave a straight line with gradient 13.5 Ohm m-¹ and a small positive intercept of 0.2 Ohm. The steepest and shallowest lines through the error bars gave gradients of 14.0 and 13.0 Ohm m-¹, so the gradient is 13.5 ± 0.5 Ohm m-¹.

Conclusion

Resistance increases linearly with length, which supports the hypothesis. Using the gradient and the area of the wire, the resistivity is (1.53 ± 0.06) × 10-^6 Ohm m. The accepted value for constantan is about 4.9 × 10-^7 Ohm m, so my value is very accurate.

Evaluation

The small positive intercept suggests a constant extra resistance, probably from the contact between the crocodile clips and the wire. This would make every resistance slightly too high but would not change the gradient. The diameter of the wire was not measured by me; I used the value printed on the reel, which may not be exact. In the future I would measure the diameter at several points with a micrometer.

Bibliography

1. Supplier data sheet for constantan wire.